

VIDYATRACK - QA TEST SCENARIOS

Format: Manual + Integration + Negative + Expected Failures

ROLE: MANAGEMENT

GLOBAL PRE-CONDITIONS (for all MANAGEMENT tests)

- Auth via OTP:
 - POST /api/v1/auth/otp/request
 - POST /api/v1/auth/otp/verify → TokenOut.access_token
- Authorization header required: Authorization: Bearer <access_token>
- Management permissions (as applicable):
 - manage_classes
 - manage_subjects
 - manage_teachers
 - manage_students
 - assign_subjects
 - register_principal
 - lock_management_access
 - read-only: attendance, marks, student_profile
- Test users:
 - M_ACTIVE (full permissions)
 - M_LIMITED (subset permissions)
 - M_INACTIVE

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1) Feature: Management Authentication & Dashboard Access
User Story: US-MGT-AUTH-01

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A) HAPPY PATH (Manual/UI)

1. Enter management phone → request OTP
2. Verify OTP
3. Management dashboard loads

B) INTEGRATION (API)

POST /api/v1/auth/otp/request
{ "phone": "9000000001" }

POST /api/v1/auth/otp/verify
{ "phone": "9000000001", "otp": "123456" }

GET /api/v1/management/dashboard
Expected 200 (ManagementDashboardOut)

C) NEGATIVE TESTS (P0)

NEG-MGT-AUTH-01.1 Inactive management user login

Expected:

- 401/403

Must NOT:

- issue access_token

NEG-MGT-AUTH-01.2 Missing token on dashboard

Expected:

- 401 Unauthorized

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2) Feature: School Academic Structure Setup

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US-MGT-01: Create and Manage Classes

A) HAPPY PATH

POST /api/v1/classes

Payload:

{ "name": "Class 6" }

Expected: 201 (ClassOut)

GET /api/v1/classes

Expected: list includes "Class 6"

B) NEGATIVE (P0)

NEG-MGT-01.1 Duplicate class name in same school

Payload:

{ "name": "Class 6" }

Expected:

- 409 / 400

Must NOT:

- create duplicate

NEG-MGT-01.2 Missing permission manage_classes

Expected:

- 403 Forbidden

US-MGT-02: Create and Manage Sections

A) HAPPY PATH

POST /api/v1/sections

Payload:

{ "class_id": 6, "name": "A" }

Expected: 201 (SectionOut)

B) NEGATIVE (P0)

NEG-MGT-02.1 Duplicate section name under same class

Expected:

- 409 / 400

NEG-MGT-02.2 Invalid class_id

Expected:

- 404 / 422

US-MGT-03: Create and Manage Subjects

A) HAPPY PATH

POST /api/v1/subjects

Payload:
{ "name": "Mathematics" }
Expected: 201 (SubjectOut)

B) NEGATIVE (P0)
NEG-MGT-03.1 Duplicate subject name
Expected:
- 409 / 400

NEG-MGT-03.2 Missing manage_subjects permission
Expected:
- 403 Forbidden

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3) Feature: Staff Registration & Lifecycle Management

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US-MGT-04: Register Teachers

A) HAPPY PATH
POST /api/v1/management/teachers
Payload:
{
 "name": "Ravi Kumar",
 "phone": "9888888888",
 "section_id": 10,
 "role": "TEACHER"
}
Expected:
- teacher created
- default status = INACTIVE

B) NEGATIVE (P0)
NEG-MGT-04.1 Duplicate phone number
Expected:
- 409 / 400

NEG-MGT-04.2 Missing manage_teachers permission
Expected:
- 403

US-MGT-05: Register Principal

A) HAPPY PATH

POST /api/v1/management/principal

Payload:

```
{  
  "name": "Suresh Rao",  
  "phone": "9777777777",  
  "email": "principal@school.com"  
}
```

Expected:

- 201
- principal visible
- only one active principal

B) NEGATIVE (P0)

NEG-MGT-05.1 Register second active principal

Expected:

- 409 / 400

NEG-MGT-05.2 Missing register_principal permission

Expected:

- 403

US-MGT-06: Activate and Deactivate Teachers

A) HAPPY PATH

- Deactivate teacher

Expected:

- teacher login blocked
- attendance/marks APIs return 403

- Reactivate teacher

Expected:

- access restored

B) NEGATIVE (P0)

NEG-MGT-06.1 Deactivate without permission

Expected:

- 403

NEG-MGT-06.2 Deactivate teacher with active session

Expected:

- in-flight API calls blocked immediately

US-MGT-07: Replace an Inactive Teacher

A) HAPPY PATH

1. Teacher A inactive
2. Teacher B active
3. Assign Teacher B to same section/subjects

Expected:

- student attendance/marks unchanged
- Teacher B can act immediately

B) DATA INTEGRITY TESTS (P0)

- Attendance history preserved
- Marks history preserved
- Student notes preserved

C) NEGATIVE

NEG-MGT-07.1 Replacement teacher inactive

Expected:

- blocked

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4) Feature: Teaching Assignment Management
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US-MGT-08: Assign Teachers to Subjects

A) HAPPY PATH

POST /api/v1/teaching-assignments

Payload:

```
{  
  "section_id": 10,  
  "subject_id": 5,  
  "teacher_id": 22  
}
```

Expected: 201 (TeachingAssignmentOut)

B) NEGATIVE (P0)

NEG-MGT-08.1 Assign inactive teacher

Expected:

- 400 / 403

NEG-MGT-08.2 Duplicate assignment

Expected:

- blocked (unique constraint)

NEG-MGT-08.3 Missing assign_subjects permission

Expected:

- 403

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5) Feature: Student Management

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US-MGT-09: Create and Manage Students

A) HAPPY PATH

POST /api/v1/students

Payload:

```
{  
  "name": "Anita",  
  "section_id": 10,  
  "roll_number": "12",  
  "parent_phone": "9666666666"  
}
```

Expected: 201 (StudentOut)

B) NEGATIVE (P0)

NEG-MGT-09.1 Missing parent_phone

Expected:

- 422

NEG-MGT-09.2 Duplicate roll number in same section/year

Expected:

- 409 / 400

NEG-MGT-09.3 Missing manage_students permission

Expected:

- 403

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6) Feature: Academic Oversight (Read-Only)

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US-MGT-10: View Academic Records

A) HAPPY PATH

GET /api/v1/attendance?date=2026-01-22

GET /api/v1/marks?section_id=10&subject_id=5&exam_type=UNIT

GET /api/v1/students/{id}

Expected:

- 200 responses
- view-only data

B) NEGATIVE (P0)

NEG-MGT-10.1 Attempt to modify attendance/marks/student

Expected:

- 403 Forbidden

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7) Feature: Parent Communication

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US-MGT-11: Send Parent Communications

A) HAPPY PATH

- Compose custom message
- Send (outbox entry created)

POST /api/v1/notifications/outbox/run

Expected:

{ "processed": <int> }

B) NEGATIVE (P0)

NEG-MGT-11.1 Attempt to change academic data via messaging

Expected:

- blocked
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8) Feature: Management Access Control & Locking

US-MGT-12: Lock Management Access

A) HAPPY PATH

- Enable lock (user with lock_management_access)

Expected:

- class/section/subject/teacher/student/assignment create blocked
- dashboards and academic views still accessible

B) NEGATIVE (P0)

NEG-MGT-12.1 User without lock_management_access tries to lock/unlock

Expected:

- 403

NEG-MGT-12.2 Attempt setup while locked

Expected:

- 403 / 409

DEFINITION OF DONE – MANAGEMENT ROLE

- Management login works via OTP
- All setup actions enforce correct permissions
- Teacher lifecycle (activate/deactivate/replace) preserves data integrity
- Academic data is view-only where specified
- Locking mechanism blocks configuration actions correctly
- No cross-school data access
- No P0/P1 defects open

END – ROLE: MANAGEMENT QA TEST SCENARIOS